

Low-Complexity Real-Time Sound Bubble Extraction for Hearables

Experiment Settings

S1. Data selection and mixture construction

The accompanying JSON gives the resolved settings for the six systems. The file lists identify the speech, noise and RIR scenes assigned to each partition. They contain identifiers, not audio or RIR arrays.

Partition	LibriSpeech subset	Speakers	Utterances
Training	train-clean-100	251	28,539
Training	train-clean-360	921	104,014
Validation	dev-clean	40	2,703
Test	test-clean	40	2,620

Only two-channel files from the corresponding WHAM training, validation and test partitions enter the ear-noise pool. The exported WHAM list applies this channel filter. The first noise channel feeds headset microphones 0–3, and the second feeds microphones 4–7.

COMSOL provides the simulated headset acoustic responses, and the image method generates the RIRs. Speech convolved with reverberant RIRs forms the mixtures; the paired direct-path RIRs provide target speech at microphone 0. Each earcup has one center microphone and three outer microphones at a nominal radius of 2.8 cm, spaced by 120°.

Nominal radius	Training scenes	Validation scenes	Query type
1 m	6,667	667	Anchor
1.5 m	6,667	667	Anchor
2 m	6,666	666	Anchor

The split seed is 20260902. Selection ranks scenes by a seeded SHA-256 key within strata, then applies the fixed quotas. Each partition contains 80% of scenes with $T_{60} \in [0.3, 0.8]$ s and 20% with $T_{60} \in (0.8, 1.2]$ s. The scene-list file records the actual assignments. Nominal radius identifies the scene pool; continuous training queries are sampled separately.

Setting	Value or rule
Mixture duration	6 s at 16 kHz; one output at microphone 0.
Simulated source segments	Uniform integer length from 2 s to the smaller of 6 s and the available utterance; shorter utterances are used in full. Crop starts and valid mixture offsets are random.
Simulated target selection	Target absent with probability 0.1; otherwise one target. At most two outside speech interferers, with distinct speakers.
Continuous query	Propose $d \sim U(0.8, 2)$ m and accept a scene-query pair only when the inside-source count matches the selected zero/one target count.
Additive ear noise	Probability 0.5 in simulation; it does not consume a speech-interferer slot.
Per-microphone perturbation	Independent gain $U(-2, 2)$ dB and integer shift uniform in $\{-2, -1, 0, 1, 2\}$ samples.
Target/interference level	Draw uniformly from $[-5, 10]$ dB when both are present. Use mean reverberant power at center microphones 0 and 4; include noise in total interference power.
Recorded source segments	Exclude the first 10 s of each speech recording, draw 2–6 s segments, and place each source independently in a 6-s mixture.
Recorded training	First-session recordings; 1-m query, 0.5-m targets, 1.75-m interferers, and target absence probability 0.3. Test speech uses the held-out second session.
Normalization	One standard deviation over every channel and sample of the mixture, shared by the mixture and all associated targets before the STFT.

S2. Network and training settings

All systems use a 320-point STFT, a 320-sample square-root Hann window and a 160-sample hop. The model-specific settings below follow the recorded training configurations. D and C denote discrete and continuous radius inputs.

ID	System	Query	Blocks	Hidden channels	LSTM units	Batch
M1	Sound Bubble, full	D	6	32	64	4
M2	Sound Bubble, small	D	6	23	23	8
M3	RC-TF-SkiMNet backbone	D	6	22	22	8
M4	Sound Bubble, small	C	6	23	23	8
M5	RC-TF-SkiMNet backbone	C	6	22	22	8
M6	Complete method	C	6	22	22	8

The small Sound Bubble systems use six blocks with 23 hidden channels and 23 LSTM hidden units, and compress frequency by a factor of 3; the full system uses no such compression. The continuous small baseline uses a 16-unit ReLU hidden layer in its radius encoder. Discrete queries use one-hot inputs for 1, 1.5 and 2 m. The baselines predict complex target spectra directly. RC-TF-SkiMNet predicts a complex mask whose real and imaginary components each lie in $[-1, 1]$ after tanh with unit scale.

RC-TF-SkiMNet module	Structure
Feature groups	RI: 16 channels; reference magnitude: 1; ILD: 13; sine/cosine IPD: 14.
Each feature encoder	Causal length-5 convolution to 22 hidden channels, LayerNorm over channels, and channel-wise PReLU. The four outputs are summed.
Radius encoder	For continuous queries, normalize d to $(d - 0.8)/1.2$. Linear $1 \rightarrow 16$, LayerNorm, PReLU, Linear $16 \rightarrow 22$, LayerNorm, PReLU. Discrete queries use a first layer of $3 \rightarrow 16$. Add before ERB compression.
ERB	Retain 30 low-frequency bins; map the remaining 131 bins to 54 fixed triangular bands. Use the transposed weights to expand features before decoding.
Local-F	1×5 convolution across frequency, SiLU and residual addition.
Full-band F	LayerNorm, frequency BiLSTM with 22 units per direction, Linear $44 \rightarrow 22$, residual addition.
Local-T	LayerNorm, causal length-5 convolution with one group, SiLU, residual addition.
Subband-T	One module after Local-T in each of blocks 1–5; none in block 6. LayerNorm, causal per-band LSTM, Linear $22 \rightarrow 22$, residual addition. Weights shared across bands; one LSTM group.
Decoder	Causal length-5 convolution from 22 to 2 channels, tanh, complex multiplication by X_0 , then inverse STFT.
VAD head, when enabled	Average final block features over frequency; LayerNorm, Linear $22 \rightarrow 22$, PReLU, Linear $22 \rightarrow 1$. Executed at inference without output gating.
Projection head	Average over time and frequency; LayerNorm, Linear $22 \rightarrow 64$, PReLU, Linear $64 \rightarrow 64$, L2 normalization. Training only.
Optimization	Shared setting
Adam	Initial learning rate 10^{-3} ; $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$, no weight decay; FP32; initial seed 7.
Epoch sampling	100,300 draws with replacement; simulation probability 0.8 and recorded-data probability 0.2. No gradient accumulation; gradient norm limited to 1.
Validation	2,000 fixed simulated examples using 1, 1.5 and 2 m anchors, one example per validation RIR. Select the lowest model-specific validation loss.
Learning-rate schedule	Plateau patience 6, reduction factor 0.5, absolute improvement threshold 0.002.
Stopping	Count non-improving epochs from epoch 60; patience 20; stop only when the learning rate is at most one quarter of its initial value. Maximum 300 epochs.

S3. Objectives and evaluation

For RC-TF-SkiMNet, the base separation objective sums a stabilized waveform SNR loss and compressed RI and magnitude errors. Let y and $\hat{y}$ be mean-centered target and estimated waveforms, $E_s = \|y\|^2$ and $E_e = \|\hat{y} - y\|^2$. No scale projection is used. The waveform loss is

$$-10\log_{10}\left(\frac{E_s + 10^{-3}}{E_e + 10^{-5}(E_s + 10^{-3})} + 10^{-5}\right).$$

For spectral compression, $m(z) = \sqrt{|z|^2 + 10^{-5}}$ and $C(z) = m(z)^{-0.7}z$. Sum the real and imaginary L1 errors of $C(\hat{Y})$ and $C(Y)$ for the RI term; use the L1 difference between $m(\hat{Y})^{0.3}$ and $m(Y)^{0.3}$ for the magnitude term. Divide each per-example spectral error by the summed $m(X_0)^{0.3}$, then average over the batch. LayerNorm uses $\varepsilon = 10^{-5}$; projection L2 normalization uses $\varepsilon = 10^{-8}$.

The Sound Bubble baselines use negative waveform SNR for nonzero targets. For an absent target, they use waveform mean L1 error weighted by 50. They do not use the second-stage multi-resolution objective.

Training term	Setting
VAD	Binary cross entropy, weight 0.1. A frame is active when its target spectral power exceeds the utterance peak minus 40 dB. Silent targets have all-zero labels.
Tail losses	Target-present SI-SDRi threshold 6 dB; absent-target suppression threshold 15 dB. Each weight is 0.05. Average the largest $\lceil 0.25N \rceil$ softplus shortfalls within each category, with temperature 1 dB.
Identity exclusion	Exclude a nonzero target from the SI-SDRi tail when input/target relative squared error is at most 10^{-12} . Keep its base and VAD supervision.
Paired queries	From V valid simulated candidates, select $P = \max(1, \text{round}(0.25V))$, with ties rounded to even. Use no second query if $V = 0$. Recorded mixtures have no second query.
Pair categories	Request different target sets with probability 0.5; otherwise request equal sets. Sample feasible intervals by length and fall back to the other category if needed.
Paired target loss	Separation plus 0.1 times VAD loss, averaged over the second queries and weighted by 0.25. No second-query tail losses.
Contrastive loss	For cosine similarity c , use $1 - c$ for equal target sets and $\max(0, c - 0.2)^2$ for different sets. Average within categories, then average the categories present.
Contrastive weight	$0.02 \min(e/3, 1)$ for epochs $e = 1, 2, \dots$. Neither the second query nor projection head is evaluated during validation or inference.

The training controls add VAD, tail losses, paired target supervision and contrastive loss in sequence. The paired-only control keeps the extra query and projection structure with contrastive weight zero. Feature controls remove ILD, IPD or ERB; the simulation-only control preserves the total epoch draws and optimizer updates.

Each domain has 100 mixtures for each of seven conditions: target only, no target, one interferer, noise, one interferer plus noise, two interferers, and two interferers plus noise. The final pair shares speech and geometry. There are 600 unique RIR scenes among the 700 standard simulated mixtures, plus 100 separate scenes for the radius test.

The target-present summary pools one interferer, noise, one interferer plus noise, and two interferers (400 mixtures per domain). Report SI-SDR improvement over microphone 0, PESQ in wideband mode and extended STOI. Report energy suppression on the 100 no-target mixtures. Low-score rates use SI-SDRi < 6 dB or suppression ≤ 15 dB, with valid case counts and separately reported invalid outputs.

Radius tests keep the single-source mixture fixed and query 1, 1.2, 1.4, 1.6, 1.8, 2 m. Each domain has 100 such mixtures. Simulated source distances lie in $[0.5, 2.5]$ m; recorded sources are at 1.75 m.

S4. Parameter and computation counts

The current measurements use ptflops 0.7.5 with the PyTorch backend and PyTorch 2.8.0 on CPU. Functional counting is disabled. Project LayerNorm subclasses use the native nn.LayerNorm counting rule. All models use an eight-microphone RI input with batch size one, 1,000 frames, 161 frequency bins, and a 10-ms hop. These frames represent 10 s of input audio. Computation is reported as MACs/($10^9 \cdot 10$) GMACs/s; it is not divided by program runtime.

Params (M) counts inference parameters in millions, including fixed ERB weights and the VAD head when present, but excluding the training-only projection. Values are rounded only for display. The JSON retains raw integer parameter and MAC counts.

System	Params (M)	GMACs/s
M1	0.50	8.24
M2	0.12	1.09
M3	0.13	1.05
M4	0.13	1.09
M5	0.13	1.05
M6	0.13	1.05

The two fixed ERB matrices contain 14,148 coefficient entries. They contribute to inference parameter counts and their matrix operations are profiled. M6 has 127,958 inference entries, including 113,810 trainable entries and the fixed ERB coefficients. Its training-only projection has 5,677 parameters and is excluded from inference counts. The VAD head is executed and retained. STFT, inverse STFT, SiLU, and unsupported functional operations are outside the module-based count.

M2 and M4 share six blocks with 23 hidden channels, 23 LSTM hidden units, and frequency compression by 3. M2 uses discrete radius inputs; M4 adds the continuous MLP16 radius encoder. Their computation is about 3.70% above M5, calculated from unrounded counts. These are comparisons at similar budgets, not identical parameter or operation counts.